d & f
Block Elements

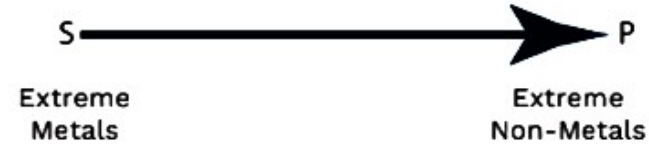
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d and f-block element

Old Definition: All d-block elements are known as transition elements.

Introduction They are often called 'transition elements' because their position in the periodic table is between s-block and p-block elements Typically, the transition elements have an incompletely filled d-level, are not considered as transition elements but they are d-block elements. Since Zn group has d^{10} configuration in their ground state as well as in stable oxidation state, they are not considered as transition elements but they are d-block elements.



Properties of Transition elements

- Variable oxidation state
- Coloured
- Paramagnetic
- Catalyst
- Interstitial Compound

Note:

Except Zn, Cd and Hg, all other elements present in d-block show these properties.

New definition of transition elements

d-block elements having variable oxidation state or having unpaired electron / partially filled d-orbital in any oxidation state are called transition elements.

$Zn : d^{10} s^2$ and $Zn^{2+} : d^{10}$

Completely filled d-orbital

According to new definition, zinc, cadmium and mercury are not transition elements but they are d-block elements. So, they are studied with transition elements.

Definitions

If a specific question is given, then zinc, cadmium and mercury are considered as transition elements but for a given (such as based on properties) zinc, cadmium and mercury are considered as transition elements.

3d series or 3d transition series

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn

Trick: Science Teacher Vihita Crari Mnn Fe Co Ni CyU Zn Jalati ho

4d series or 4d transition series

Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd

Trick: Yaari Zra Nibha Mout Tc Rukawat Rh Pdi Ag mein Cd Jaa

General Electronic Configuration

- In general, the electronic configuration of these elements is $(n-1)d^{1-10}ns^{1-2}$.
- The (n–1) stands for the inner shell and the d-orbitals may have one to ten electrons and the s-orbital of the outermost shell (n) may have one or two electrons.

Points to remember

Exception of electronic configuration in d- block

Nb Cu Cr Rg Mo Tc Ru Rh Pd Au Ag Pt

1 electron difference from $n+\ell$ — $2e^-$ difference from $n+\ell$

Tips and Trick

Navabi kyu kare raga maut tak rukawate rah mai padegi aur pitaya hogi

Example :

Ni $3d^8$ $4s^2$

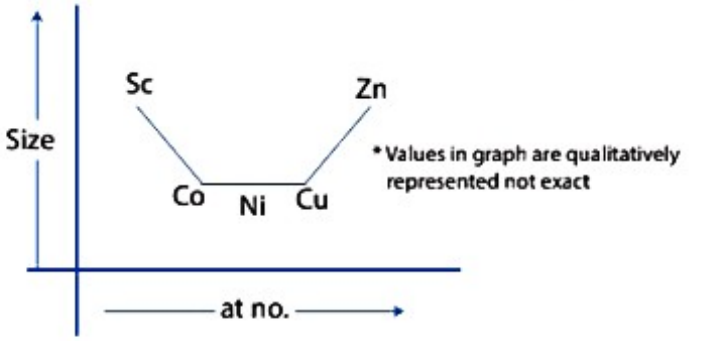
Pd $4d^{10}$ ---------- $2e^-$ difference from $(n+\ell)$

Pt $5s^9$ $6s^1$---------- $1e^-$ difference from $(n+\ell)$

Physical properties

(1) Atomic size

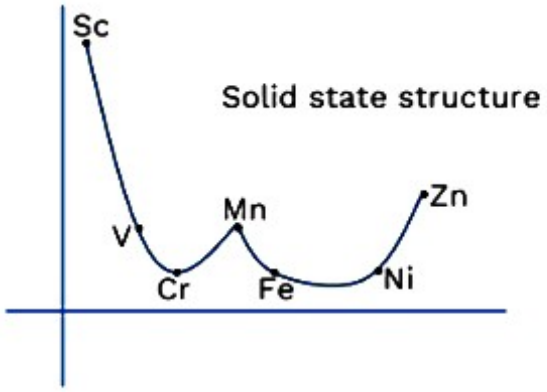
Sc Ti V Cr	Mn Fe Co Ni Cu Zn
Unpaired e^- ↑	Unpaired e^- ↓ | d^{10}
Metal-Metal bonding increases	Metal–Metal Bonding ↓
Attraction↑	Attraction↓
Radius↓	Radius↑
Or size↓	Or size ↑



Sc → Mn decreases | Fe | Co — Ni — Cu Remains same | Zn Increases again

(2) Metallic Radius

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
164	147	135	129	137	126	125	125	128	137



$3d < 4d \approx 5d$

Due to poor shielding of 4f electrons that's lanthanoid contraction.

(3) Ionic Radius

Depends on z_{eff}

-Z_{eff} ↑, size↓

(4) Solid State Structure

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
HCP (BCC)	BCC (HCP)	BCC	BCC	×→not defined HCP	HCP	CPP			

(5) Melting & Boiling Point

Melting point:
3d series

Exception (∵it has solid state structure)

Sc Ti V Cr

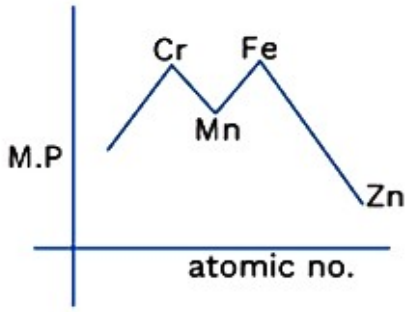
Unpaired↑
M-M bonding↑
attraction↑
Melting point↑

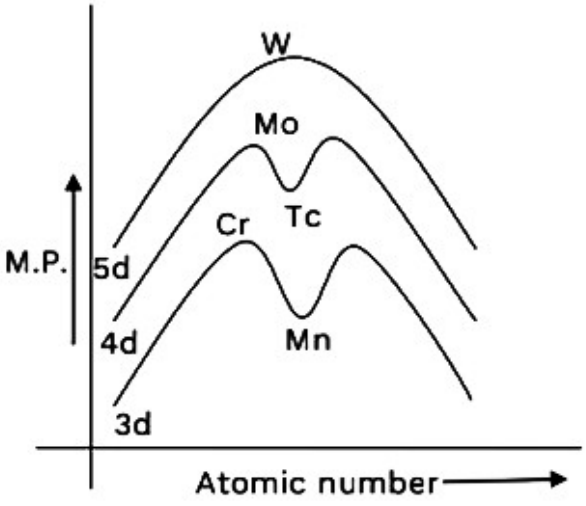
Mn Fe Co Ni Cu Zn

Unpaired electron↓
M-M bonding↓
Melting point↓

Cr, Mo, W → Maximum MP
6 no. of unpaired e⁻s
Are involved in metallic bonding

Zn, Cd, Hg → Lowest MP
Due to no unpaired e⁻
For metallic Bonding





- Zinc has minimum melting point in 3d series of transitional elements

Points to remember

Order of melting point: 3d<4d<5d
Reason:

- Z_{eff} ↑
- M-M bonding ↑
- Attraction ↑
- Melting point ↑

Tips and Trick

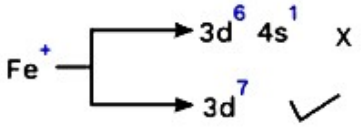
Exception:
Size: Cu < Ag ≃ Au
M.P Cu > Ag < Au
or Cu > Au > Ag
Example: Zn > Cd > Hg (liquid)

(6) Density

Order of density: 3d < 4d < 5d

- As we move down the group, mass increases rapidly as compared to size. So, density increases.
- As we move along the period, density increases because size decreases and remains almost same but mass increases.
- Configuration of transition metal of M^{+1} type.

Example:



Fe : $3d^6 4s^2$

- After removal of one e^- from Fe, it becomes Fe^+

Result

Positive charge increases, attraction on an outer electron also increases ⇒ it moves in (n – 1) d.
i.e., it becomes ns°.

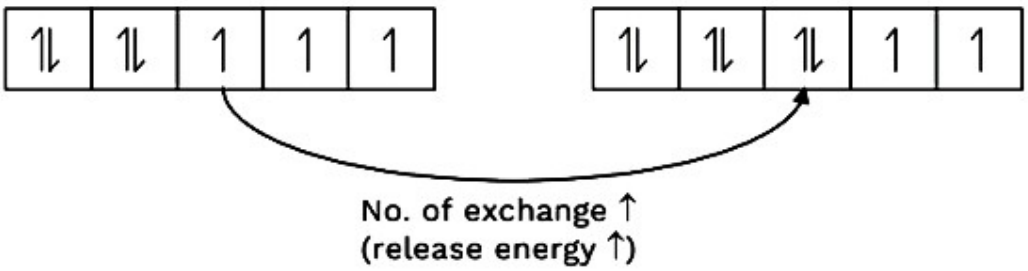
(7) Ionization energy of 3d series

	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
I.E.	631	656	650	653	717	762	758	736	745	906

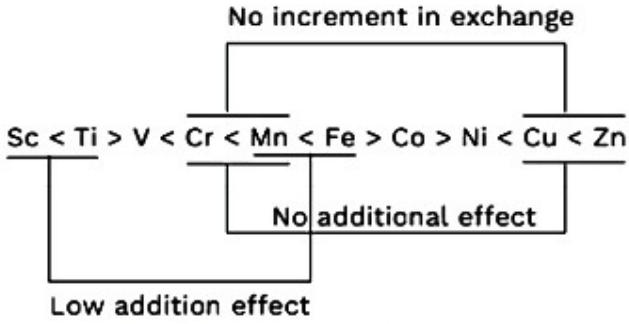
Order of Ionization Energy
Sc < Ti > V < Cr < Mn < Fe > Co > Ni < Cu < Zn

Example :

$$Co_{3d^7 4s^2} \xrightarrow[\text{energy required}]{-e^-} Co^+_{3d^8}$$



- In general left to right Zeff ↑ so I.E ↑ ses.
- Additional effect in observed due to increment in number of exchange due to shifting of electron from ns to (n – 1)d
- Due to additional effect in some cases decrement in I.E. observed.



- 1st, 2nd, 3rd, Ionization energies are increasing from left to right for 1st Transition series, but not regularly.
- For 2nd IE Cr > Fe > Mn and Cu > Zn
- For 3rd IE Mn > Cr > Fe and Z has highest.

(8) Variable Oxidation State : Oxidation state of the Ist transition series Most common ones are in below types:

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
			+1					+1	
	+2	+2	+2	+2	+2	+2	+2	+2	+2
+3	+3	+3	+3	+3	+3	+3	+3		
	+4	+4	+4	+4	+4	+4	+4		
		+5	+5	+5					
			+6	+6	+6				
				+7					

Maximum oxidation state = ns electron + (n – 1)d electron unpaired. The elements which give the greatest number of oxidation states occur in or near the middle of the series. Cu^{+1} is stable only in solid state. $Cu^{+}(aq.) \rightarrow Cu^{2+} + Cu$	Maximum oxidation state is achieved with oxygen and fluorine. They also show low oxidation state with CO and π -acid ligands due to synergic bonding. $3d \rightarrow 4d \rightarrow 5d$ $Z_{eff}\uparrow$, splitting energy$\uparrow$ stability$\uparrow$

Example:

(i) Stability $Cr^{+6} < Mo^{+6} < W^{+6}$

(ii) Among chromium, molybdenum and tungsten, which is good oxidising agent.

Ans. Chromium.

Points to remember

Stability of highest oxidation state in d-block increases as we move down the group

Trends in stability of higher oxidation state :

(1) Table shows the stable halides of the 3d series of transition metals. The highest oxidation numbers are achieved in TiX_4 (tetrahalides), VF_5 and CrF_6. The +7 state for Mn is not represented in simple halides but MnO_3F is known, and beyond Mn no metal has a trihalide except FeX_3 and CoF_3.

(2) The ability of fluorine to stabilise the highest oxidation state is due to either higher lattice energy as in the case of CoF_3, or higher bond enthalpy terms for the higher covalent compounds, Example :, VF_5 and CrF_6.

(3) Although V^{v} is represented only by VF_5, the other halides, however, undergo hydrolysis to give oxohalides, VOX_3.

(4) Another feature of fluorides is their instability in the low oxidation states Example: VX_2 (X = Cl, Br or I)

Formulas of halides of 3d-metals

Oxidation Number									
+6			CrF_6						
+5		VF_5	CrF_5						
+4	TiX_4	VX_4^I	CrF_4	MnF_4					
+3	$TiX_{3_{III}}$	VX_3	CrF_3	MnF_3	FeX_3^I	CoF_3		CuX_2^{II}	ZnX_2
+2	TiX_2	VX_2	CrF_2	MnX_2	FeX_2	CoX	NiX_2	CuX^{III}	
+1									

(9) E° (Standard Electrode Potential)

According to concept of thermodynamics spontaneity of any process or reaction define by change in Gibbs free energy (ΔG).

1. If ΔG = negative then reaction is spontaneous.
2. If ΔG = positive then reaction is non-spontaneous.

$$\underset{-ve}{\underline{\Delta G}} = -nf \underset{+ve}{\underline{E}}$$

Tips and Trick

Oxidation state relatable with colour :

$Ta^{4+} \rightarrow$ colourless
$V^{4+} \rightarrow$ blue
$V^{2+} \rightarrow$ violet
$Cr^{3+} \rightarrow$ green
$Mn^{2+} \rightarrow$ light pink
$Fe^{3+} \rightarrow$ yellow
$Ni^{2+} \rightarrow$ green
$Zn^{2+} \rightarrow$ colourless

$Sc^{3+} \rightarrow$ colourless
$Ti^{3+} \rightarrow$ purple
$V^{3+} \rightarrow$ green
$Cr^{2+} \rightarrow$ blue
$Mn^{3+} \rightarrow$ violet
$Fe^{2+} \rightarrow$ light green
$Co^{2+} \rightarrow$ pink
$Cu^{2+} \rightarrow$ blue

Now for electrode potential we have to understand the process of conversation of solid metal to metal ion in aqueous medium with the help of given cycle

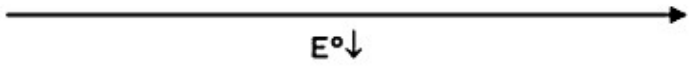
M^{+2}/M $\quad$ $M^{+2} \rightarrow M$

M/M^{+2} $\quad$ $M(s) \longrightarrow M^{+2}(aq)$

$$M(g) \xrightarrow[IE_1+IE_2]{} M^{+2}(aq)$$

So variation of electrode potential (E°) for 3-d series according to NCERT

Ti	V	Cr	Mn	Fe	Co	Ni	Cu
−1.63	1.18	0.90	1.18	0.44	0.28	0.25	−0.34



E°↓

As going left to right in period electrode potential (E°) decreases

And for M / M^{+2} : E° = +ve then Reactivity > H_2

M / M^{+2} : E° = –ve then Reactivity < H_2

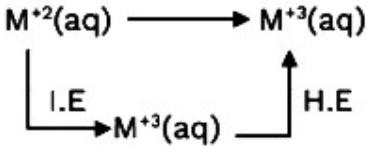
Reactivity Series

Li	Mg	Co	Ag
K	Al	Ni	Au
Ba	Mn	Sn	Pt
Sr	Zn	Pb	
Ca	Cr	H_2	
Na	Fe	Cu	
	Cd	Hg	

By above explanation and according to electro chemical series we can say that:

- $Cu \rightarrow Cu^{2+} + 2e^-$ is non spontaneous ; As we move from left to right, standard electrode potential E°↓, means reactivity in water ↓, so Z_{eff} (effective nuclear charge)↑ and I.E.(ionisation energy)↑ and releasing tendency of electron decreases and electrode potential (E°) decreases.
- For $Mn^{2+}\left(d^5\right)$, to achieve half filled configuration IE_2 of Mn should be less from its expected value. So to attain d5 (half filled configuration) , reactivity will increase.

  Now formation of M^{+2} to M^{+3} ;



Two possibilities for Hydration and Ionisation

I.E > H.E	I.E < H.E
Required > release $E°_{M+2/M+3}$ = -ve i.e., Non spontaneous M^{+2} is more stable than M^{+3}	Required < Release $E°_{M^{+2}/M^{+3}}$ = +ve i.e. Spontaneous M^{+3} is more stable than M^{+2}

So according to given data in NCERT value of electrode potential

Ti	V	Cr	Mn	Fe	Co
+0.37	0.26	0.41	–1.57	–0.77	–1.97

Stability $M^{+2} < M^{+3}$ $M^{+2} > M^{+3}$

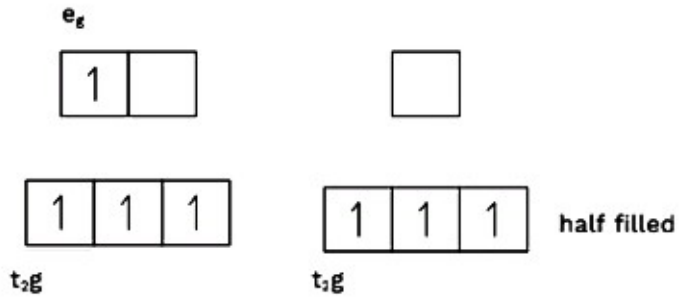
As we move along the period, Zeff. ↑, I.E↑ and E°↓ except chromium and Iron in 3d series

Example :

$Fe^{2+} \rightarrow Fe^{3+}(d^5$ half filled)

$Cr^{2+}_{(aq)} \rightarrow Cr^{3+}$

d^4



Stability of +3 in Chromium, Manganese and Iron: $Cr^{+3} > Fe^{+3} > Mn^{2+}$

Element	Sc		Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Atomic number	21		22	23	24	25	26	27	28	29	30
Electronic configuration	M	$3d^1$ $4s^2$	$3d^2$ $4s^2$	$3d^3$ $4s^2$	$3d^5$ $4s^2$	$3d^5$ $4s^2$	$3d^6$ $4s^2$	$3d^7$ $4s^2$	$3d^8$ $4s^2$	$3d^{10}$ $4s^1$	$3d^{10}$ $4s^2$
	M^+	$3d^1$ $4s^1$	$3d^2$ $4s^1$	$3d^3$ $4s^1$	$3d^5$	$3d^5$ $4s^1$	$3d^6$ $4s^1$	$3d^7$ $4s^1$	$3d^8$ $4s^1$	$3d^{10}$	$3d^{10}$ $4s^1$
	M^{2+}	$3d^1$	$3d^2$	$3d^3$	$3d^1$	$3d^5$	$3d^6$	$3d^7$	$3d^8$	$3d^9$	$3d^{10}$
	M^{3+}	[Ar]	$3d^1$	$3d^2$	$3d^3$	$3d^1$	$3d^5$	$3d^6$	$3d^7$	-	-
Enthalpy of atomization $\Delta_1H^{\ominus}$/kJ mol^{-1}		326	473	515	397	281	416	425	430	339	126
Ionisation Enthalpy, $\Delta_1H^{\ominus}$/kJ mol^{-1}	I	631	656	650	653	717	762	758	736	745	906
$\Delta_1H^{\ominus}$	II	1235	1309	1414	1592	1509	1561	1644	1752	1958	1734
	III	2393	2657	2833	2990	3260	2962	3243	3402	3556	3829

(10) Magnetic Behaviour

Spin only magnetic moment, $\mu = \sqrt{n(n+2)}$B.M

One unpaired means ⇒1BM
Two unpaired ⇒2BM
Three unpaired ⇒ 3 BM
where n = number of unpaired electrons.

Tips and Trick

d^{1-5} ⇒ expected = experimental
d^{6-9} ⇒ expected < experimental
(i.e. spin + orbital)

Q1 **The magnetic moment (B.M.) of $[Fe(CN)_6]^{4-}$ & $[Fe(H_2O)_6]^{2+}$ Cl^- respectively are-**

(A) $\sqrt{24}$ **, zero** **(B)** $\sqrt{24}$ **,** $\sqrt{24}$
(C) Zero, $\sqrt{24}$ **(D) zero, zero**

Sol. C

F_e^{2+} n = 0 , MM = 0

Fe^{2+} n = 4 MM = $\sqrt{24}$

Q2 **Arrange the following in increasing order of their magnetic moment**

$[Fe(CN)_6]^{4-}$, $[Fe(CN)_6]^{3-}$, $[Ni(CN)_4]^{2-}$ & $[Ni(H_2O)_4]^{2+}$

(A) $[Fe(CN)_6]^{4-}$, $[Ni(CN)_4]^{2-}$, $[Ni(H_2O)_4]^{2+}$, $[Fe(CN)_6]^{3-}$

(B) $[Ni(H_2O)_4]^{2+}$, $[Ni(CN)_4]^{2-}$, $[Fe(CN)_6]^{4-}$, $[Fe(CN)_6]^{3-}$

(C) $[Fe(CN)_6]^{3-}$, $[Fe(CN)_6]^{4-}$, $[Ni(CN)_4]^{2-}$, $[Ni(H_2O)_4]^{2+}$

(D) $[Fe(CN)_6]^{4-}$ = $[Ni(CN)_4]^{2-}$, $[Fe(CN)_6]^{3-}$, $[Ni(H_2O)_4]^{2+}$

Sol. D

$[Fe(CN)_6]^{4-}$ n =0 MM = 0
$[Ni(CN)_4]^{2-}$ n = 0 MM = 0
$[Fe(CN)_6]^{3-}$ n = 1 MM = $\sqrt{2}$
$[Ni(H_2O)_4]^{2+}$ n = 2 MM = $\sqrt{8}$

Q3 **What will be the charge on Fe^{x+} if the magnetic moment is $\sqrt{24}$ -**

(A) + 2 **(B) + 3**

(C) Zero **(D) None of these**

Sol. A

n = 4 $\therefore$ Fe^{2+}

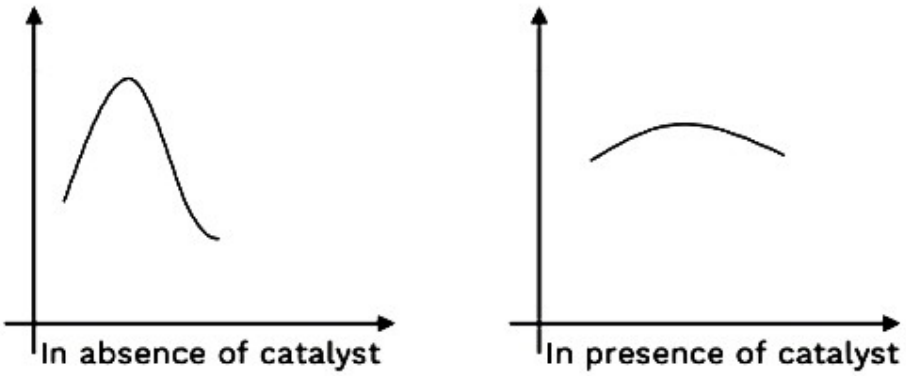
If BM = $\sqrt{24} = \sqrt{4(4+2)}$

number of unpaired e = 4

Then Fe must have +2 charge

(11) Catalytic Property

- Catalyst provide new and better path for reaction, with low energy barrier.



- Transition metals are used as catalyst due to variable oxidation state and active surface valency.
- The transition metals and their compounds are known for their catalytic activity. This activity is described to their ability to adopt multiple oxidation states and to form complexes. Vanadium (V) oxide (in Contact Process), finely divided iron (in Haber's Process), and nickel (in Catalytic Hydrogenation) are some of the examples.
- Catalysts at a solid surface involve the formation of bonds between reactant molecules and atoms of the surface of the catalyst (first row transition metals utilise 3d and 4s electrons for bonding).
- This has the effect of increasing the concentration of the reactants at the catalyst surface and also weakening of the bonds in the reacting molecules (the activation energy is lowering).

Example:

(i) $N_2 + H_2 \rightarrow$ No reaction (due to high bond dissociation energy of N_2)

$$N_2 + 3H_2 \xrightarrow{Fe(Catalyst)} 2NH_3$$

Electron from Fe enters in antibonding molecular orbital of N_2 and H_2. So, bond dissociation energy decreases.

(ii) $SO_2 + O_2 \rightarrow$ No reaction ($\because$ Activation energy is high)

$$2SO_2 + O_2 \xrightarrow{V_2O_5} 2SO_3$$

$$\left[\begin{array}{c} SO_2 + V_2O_5 \rightarrow SO_3 + 2VO_2 \\ 2VO_2 + \frac{1}{2}O_2 \rightarrow V_2O_5 \end{array}\right]$$

In V_2O_5, vanadium show variable oxidation state. That's why, V_2O_5 act as catalyst.

Special case : Also because the transition metal ions can change their oxidation states, they become more effective as catalysts.

Example:

Iron(III) catalyses the reaction between iodide and persulphate ions.

$2I^- + S_2O_8^{2-} \rightarrow I_2 + 2SO_4^{2-}$

An explanation of this catalytic action can be given as:

$2Fe^{3+} + 2I^- \rightarrow 2\ Fe^{2+} + I_2$

$2\ Fe^{2+} + S_2O_8^{2-} \rightarrow 2Fe^{3+} + 2SO_4^{2-}$

Catalyst		Used	
(i)	$TiCl_3$	(a)	Used as the Ziegler-Natta catalyst in the production of polythene.
(ii)	V_2O_5	(b)	Convert SO_2 to SO_3 in the contact process for making H_2SO_4
(iii)	MnO_2	(c)	Used as a catalyst to decompose $KClO_3$ to give O_2
(iv)	Fe	(d)	Promoted iron is used in the Haber-Bosch process for making NH_3
(v)	$PdCl_2$	(e)	Wacker process for converting $C_2H_4 + H_2O + PdCl_2$ to $CH_3CHO + 2HCl + Pd$.

(vi)	Pd	(f)	Used for hydrogenation (e.g. phenol to cyclohexanone).
(vii)	Pt/PtO	(g)	Adams catalyst, used for reductions.
(viii)	Pt	(h)	Formerly used for $SO_2 \rightarrow SO_3$ in in the contact process for making H_2SO_4
(ix)	Pt/Rh	(i)	Formerly used in the ostwald process for making HNO_3 to oxidize NH_3 to NO
(x)	Cu	(j)	Is used in the direct process for manufacture of $(CH_3)_2SiCl_2$ used to make silicones.
(xi)	$CuCl_2$	(k)	Decon process of making Cl_2 from HCl
(xii)	Ni	(l)	Raney nickel, numerous reduction processes (e.g. manufacture of hexamethylenediamine, production of H_2 from NH_3, reducing anthraquinone to anthraquinol in the production of H_2O_2
(xiii)	$FeSO_4$ + H_2O_2	(m)	Used as Fenton's reagent for oxidizing alcohols to aldehydes.

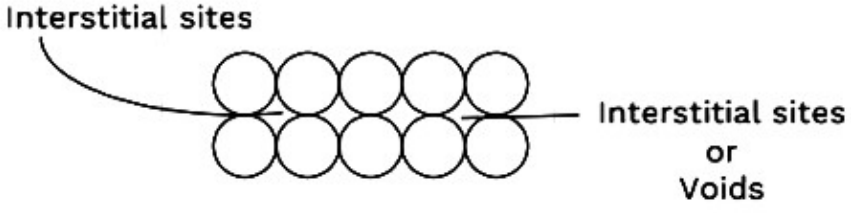
(12) Interstitial compounds or Non stoichiometric Compound

Interstitial compounds are those which are formed when small atoms like H, C or N are trapped inside the crystal lattices of metals. The principal physical and chemical characteristics of these compounds are as follows:

Points to remember

Non-stoichiometric means not in simple ratio

(i) They have high melting points, higher than those of pure metals.
(ii) They are very hard, some borides approach diamond in hardness.
(iii) They retain metallic conductivity.
(iv) They are chemically inert.



- Reactivity of d-block metals is less
- When small size H/C/B/N atoms enter in voids of d-block metal, they from interstitial compounds.
- They are not simple compounds
- Their formula is not in simple ratio.
- There is no bond between metallic atoms and small size atoms
- There is no change in number of free electrons
- Electrical conductivity remains same
- Hardness ↑, Density ↑, M.P.↑
- Ductility and malleability decreases due to presence of small size atom.

(13) Colour of ions in aqueous medium

aqueous medium → "Ligand" = H_2O

Ions make octahedral aqua complex having coordination number 6 .Compounds of transition metals are usually coloured. Colour is associated with the ability of a compound to promote an electron from one energy level to another higher energy level by absorbing light of particular wavelength. In transition elements, d-d transitions are responsible for colour. In d^0 and d^{10} compounds, it is not possible, therefore, to promote electrons within the d level.

i.e.

d^0 → empty

d^{10} → full

d^5 – WFL → generally coloured due to LMCT (Ligand to metal charge transfer)

where WFL = Weak field ligand

d^0, d^{10} → colourless

d^1, d^9 → Blue

d^2, d^8 → Green

Example :

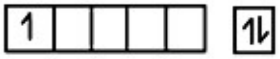
Ions	Colour
Cu^{2+},Cr^{2+}	Blue
Mn^{2+},Co^{2+}	Pink
Fe^{2+},Ni^{2+}	Green
Cr^{3+}	Violet
Fe^{3+}	Pale yellow

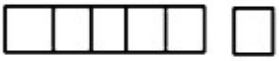
Hydrated Cu^{2+} ions and anhydrous cobalt (II) compounds absorb radiations of wavelengths corresponding to red light. $(\lambda = 605 - 765nm)$ and appear, therefore, blue.

Q4 **Which of the following ions are colourless-**

(A) V^{3+} **(B)** Mn^{2+} **(C)** Sc^{3+} **(D)** Ti^{2+}

Sol. C

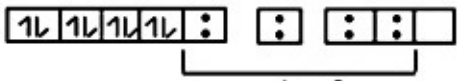
Sc(21) 

Sc^{+3}  Colorless

Q5 **Match the geometry with the complex ion -**

	X		Y
(a) Octahedral			**(I)** $[Ni(CN)_4]^{2-}$
(b) Square Planer			**(II)** $[Fe(CN)_6]^{4-}$
(c) Tetrahedral			**(III)** $Ni(CO)_4$
	a	**b**	**c**
(A)	I	II	III
(B)	II	III	I
(C)	II	I	III
(D)	I	III	II

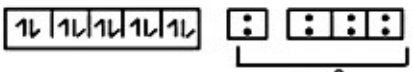
Sol. C

$Ni(CN)_4$ 

dsp^2 square planar

$[Fe(CN)_6)^{-4}$ 

d^2sp^3 octahedral

$Ni(Co)_4$ 

sp^3 tetrahedral

Q6 **AgCl and NaCl are colourless NaBr & NaI are also colourless but AgBr and AgI are coloured. This is due to -**

(A) Ag^+ has half filled d- orbital **(B) Ag^+ polarises Br^- & I^-**

(C) Ag^+ depolarises Br^- & I^- **(D) None is correct**

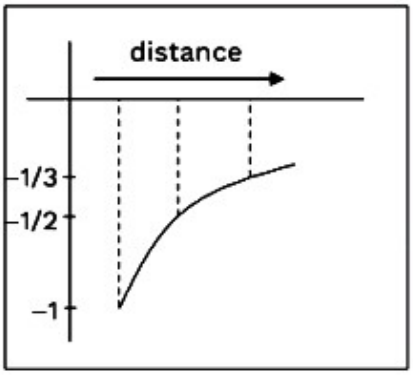
Sol. B

AgCl NaCl, NaBr, NaI less polarization, more ionic, colorless AgBr, AgI more polarisation, less ionic, more covalent, colored

(14) Complex formation tendency

Complex formation i.e., (Lewis acid-Base reaction)

Complex compounds are those compounds in which the metal ions bind a number of anions or neutral molecules giving complex species with characteristic properties. The transition metals form a large number of complex compounds. This is due to small sizes of metal ions, have vacant orbital, more effective nuclear charge and exhibit positive charge.



For

$$E = \frac{Kq_1q_2}{r}$$

r = 1 E = –1

r = 2 E = –1/2

r = 3 E = –1/3

Whenever there is force of attraction between particles, energy release increases, attraction increases, distance decreases and itself energy decreases.

Q7 **The ability of d- block elements to form complexes is due to -**

(A) small and highly charged ions

(B) Vacant low energy orbital to accept lone pair of electrons from ligands

(C) Both A & B are correct

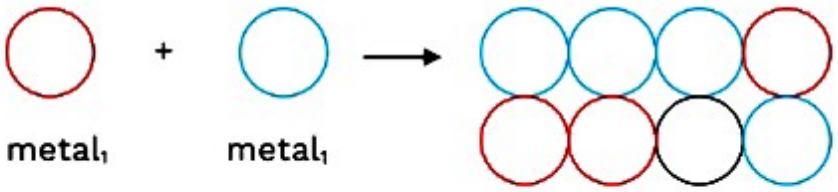
(D) None is correct

Sol. C

The ability of d-block element to form is due to the small and highly charged ions & vacant low energy orbital to accept lone pair electrons from ligands

(15) Alloy formation tendency

An alloy is a mixture of metals or alloy is a blend of metals prepared by mixing the components. Alloys may be homogeneous solid solutions in which the atoms of one metal are distributed randomly among the atoms of the other. Such alloys are formed by atoms with metallic radii that are within about 15 percent of each other. Because of similar radii and other characteristics of transition metals, alloys are readily formed by these metals. The alloys so formed are hard and have often high melting points. The best known are ferrous alloys: chromium, vanadium, tungsten, molybdenum and manganese are used for the production of a variety of steels and stainless steel. Alloys of transition metals with non transition metals such as brass (copper-zinc) and bronze (copper-tin), are also of considerable industrial importance.



Atoms having size difference less than 15% are used to make alloy. The d-block metals have almost same size so they are used to make alloy.

Example :

Brass (Copper-zinc and Bronze copper-tin) Hg alloy (amalgam), Fe alloy (steel).

(16) Halides of 3d series

	Ti	V	Cr	Mn	Fe	Co	Ni	Cu
+6			CrF_6					
+5		VF_5	CrF_5					
+4	TiX_4	VX'_4	CrX_4	MnF_4				
+3	TiX_3	VX_3	CrX_3	MnF_3	FeX'_3	CoF_3		
+2	TiX''_2	VX_2	CrX_2	MnX_2	FeX_2	CoX_2	NiX_2	CuX_2'' CuX'''
+1								

Where X = F, Cl, Br, I

X^{I} = F, Cl, Br

X^{II} = F, Cl

X^{III} = Cl, Br, I

Note:

$FeI_3, CuI_2, CuBr_2, TiF_2, VI_4, CuF$ do not exist.

Explanation:

- FeI_3 , CuI_2 , $CuBr_2$, VI_4 do not exist due to I^- is reducing nature.
- In maximum, oxidation state, only F^- exist. Example :, CrF_6, VF_5
- MnF_7, MnF_6 and MnF_5 do not exist due to steric crowding.
- TiF_2 and CuF do not exist due to oxidising nature of 'F'
- So many bromide (Br^-) and iodide (I^-) do not exist due to reducing nature.

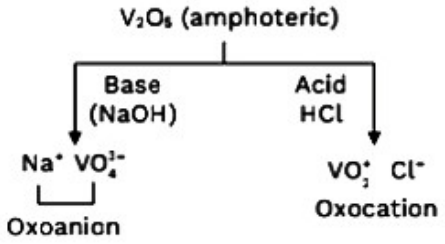
(17) Oxides

Oxidation No	3	4	5	6	7	8	9	10	11	12
+7					Mn_2O_7					
+6				CrO_3						
+5			V_2O_5							
+4		TiO_2	V_2O_4	CrO_2	MnO_2					
+3	Sc_2O_3	Ti_2O_3	V_2O_3	Cr_2O_3	Mn_2O_3 Mn_3O_4	Fe_2O_3 Fe_3O_4	Co_3O_4			
+2		TiO	VO	(CrO)	MnO	FeO	CoO	NiO	CuO	ZnO
+1									Cu_2O	

Explanation:

- CrO_3, Mn_2O_7 are acidic oxide.
- TiO_2, V_2O_5, CrO_2, Cr_2O_3 and MnO_2 are amphoteric –oxides.
- Remaining oxides of transition elements are basic in nature.
- Oxo-cations stabilise V^{+5} as VO_2^+, V^{+4} as VO^{2+} and Ti^{IV} as TiO^{2+} (because they can not exist in free form due to high polarising power).

For example: when V_2O_5 react with acid or base then in case of acid it will exist in form of oxa-cation and in case of base it will exist in form of oxa-anion.



(18) Potassium Dichromate, $K_2Cr_2O_7$:

Preparation:

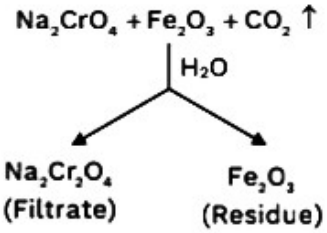
Different steps involved in the preparation of potassium from chromite ore:

(a) Preparation of Sodium Chromate:

Finely powdered chromite ore is mixed with soda ash and quick lime and then roasted in rotating furnace in excess of air. The mass turns yellow due to formation of Sodium chromate.

$$4Fe(CrO_2)_2 + 8Na_2CO_3 \xrightarrow[(O_2)]{\text{Fused}} 8Na_2CrO_4 + 2Fe_2O_3 + 8CO_2 \uparrow$$

(b) The yellow mass is extracted with water when the ferric oxide is left behind as precipitate.



(c) Conversion of chromate into dichromate

Sodium chromate solution obtained is treated with conc. sulphuric acid when it is converted into sodium dichromate.

$$2Na_2CrO_4 + H_2SO_4 \longrightarrow Na_2Cr_2O_7 + Na_2SO_4 + H_2O$$

(d) Conversion of sodium dichromate into potassium dichromate:

Hot concentrated solution of sodium dichromate is treated with potassium chloride when potassium dichromate, being the less soluble, crystallizes out on cooling.

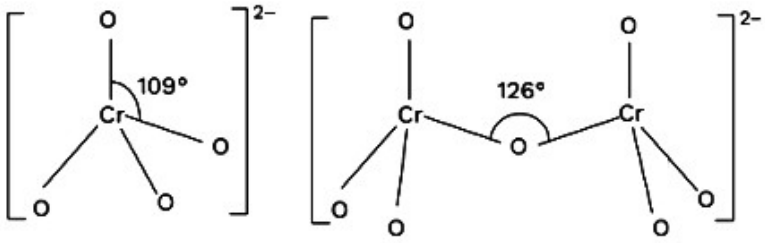
$$\underset{(K_i > K_{sp})}{Na_2Cr_2O_7} + 2KCl \longrightarrow \underset{(K_{sp})}{K_2Cr_2O_7} + 2NaCl$$

Explanation: For above reaction if K_{sp} (solubility product) is less then K_i (ionic product) than precipitate will form.

$K_{sp} > K_{sp'}$
$K_{sp'} < K_i$ means ppt. formation

Note:
LMCT is Ligand to metal charge transfer.

Structure of $K_2Cr_2O_7$:



Chromate d°

Dichromate d°

1. Charge transfer	LMCT	LMCT
2. Colour	Yellow	Orange
3. Geometry	Tetrahedral	Tetrahedral (bridge)

Properties

Points to remembered

- $Na_2Cr_2O_7$ is used as oxidising agent in organic reaction.
- $Na_2Cr_2O_7$ is hygroscopic (water absorbers) in nature. So, it is not used as primary standard (compound which is used as reference in titration to identify concentration of unknown) in redox titration.
- $K_2Cr_2O_7$ is not hygroscopic, so, it is used us primary standard.
- $Cr_2O_7^{-2}$ act as oxidising agent only in acidic medium [as $[H^+]\uparrow$, oxidising nature increases)
- In acidification of $K_2Cr_2O_7$, only dilute H_2SO_4 is used.
  Because

$$\begin{bmatrix} \text{conc. } H_2SO_4 \rightarrow \text{dehydrating agent} \\ \text{dil/conc.} HNO_3 \rightarrow \text{oxidising agent} \\ HCl/HBr/HI \rightarrow \text{Reducing agent} \end{bmatrix}$$

So these cannot use for acidification.

Acidification reaction:

$$K_2Cr_2O_7 + 4H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + 3O$$

$$Cr_2O_7^{-2} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$$

So $Cr_2O_7^{-2}$ act as a oxidising agent.

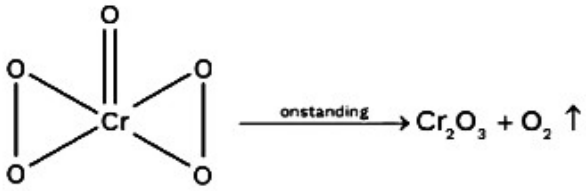
Example:

So $K_2Cr_2O_7$ use for oxidation following and reaction involve in oxidation.

$$K_2Cr_2O_7 + \underset{\text{(Acidified)}}{} \left[\begin{array}{l} \text{(a) } HCl \rightarrow Cl_2 \\ \text{(b) } HBr \rightarrow Br_2 \\ \text{(c) } HI \rightarrow I_2/I_3^- \\ \text{(d) } HNO_2 \rightarrow HNO_3 \\ \text{(e) } H_2SO_3 \rightarrow H_2SO_4 \\ \text{(f) } H_2C_2O_4 \rightarrow CO_2 \\ \text{(g) } Fe^{2+} \longrightarrow Fe^{3+} \\ \text{(h) } Sn^{2+} \longrightarrow Sn^{4+} \\ \text{(i) } H_2S \longrightarrow S \\ \text{(j) } S \longrightarrow SO_2 \\ \text{(k) } ClO_2 \longrightarrow \text{No reaction} \end{array}\right] + Cr^{+3}_{(aq.)}$$

- $K_2Cr_2O_7(s) + \underset{(conc.)}{H_2SO_4} + KCl \rightarrow \underset{\text{Chromyl chloride}}{CrO_2Cl_2} \xrightarrow{NaOH} Na_2CrO_4(aq.) \xrightarrow{Pb^{2+}(aq.)} PbCrO_4 \downarrow$

$$Cr^{3+} \xrightarrow{H_2O_2/OH^-} \overset{+6}{Cr}O_4^{2-} \xrightarrow{H_2O_2/H^+} CrO_5\,(\text{Blue})$$



Q8 **In laboratory $K_2Cr_2O_7$ is used mainly not $Na_2Cr_2O_7$. Why?**

Sol. $Na_2Cr_2O_7$ is deliquescent enough and changes its concentration and can not be taken as primary standard solution whereas $K_2Cr_2O_7$ has no water of crystallisation and not deliquescent.

Q9 **How to standardise $Na_2S_2O_3$ solution in iodometry?**

Sol. $K_2Cr_2O_7$ is primary standard ⇒ strength is known by weighing the salt in chemical balance and dissolving in measured amount of water.
Then in acidic solution add. KI

$Cr_2O_7^{2-} + 14H^+ + 6I^- \rightarrow 2Cr^{3+} + 3I_2 + 7H_2O$

This I_2 is liberated can be estimated with $S_2O_3^{2-}$.

(19) Potassium Permanganate, $KMnO_4$:

Preparation:

1. $2MnO_4 + 4KOH + (air/KNO_2/KClO_3) \xrightarrow{fused} 2K_2MnO_4 + H_2O$
   Potassium permanganate is prepared by fusing manganese dioxide with caustic potash or potassium carbonate in presence of atmospheric oxygen or any other oxidising agent (Example: potassium nitrate). The fused mass is green due to formation of potassium manganate.
2. $3MnO_4^{2-} + 2H_2O \longrightarrow 2MnO_4^- + MnO_2 \downarrow + 4OH^-$
3. $2MnO_4^{2-} + Cl_2 \longrightarrow 2MnO_4^- + 2Cl^-$
4. $2MnO_4^{2-} + O_3 + H_2O \longrightarrow 2MnO_4^- + 2OH^- + O_2$

Explanation: $KMnO_4$ is stable in basic medium but unstable in acidic medium So KOH is used but after formation of K_2MnO_4 above reaction takes place in various medium.

- **In reaction 1**, manganese is oxidised (basic medium) to from K_2MnO_4.

- **In reaction 2**, medium is acidic and neutral, so it is unstable and disproportionate reaction takes place. This reaction is not used for preparation of $KMnO_4$ because in addition to $KMnO_4$, MnO_2 is also formed.

  $\rightarrow Mn^{2+} \xrightarrow{PbO_2/S_2O_8^{2-}/Bi^{+5}} MnO_4^-$

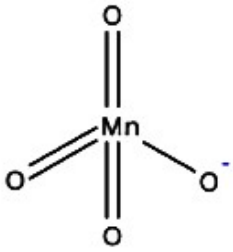
  This reaction is also not used in preparation of $KMnO_4$ because due to presence of impurity of inter mediate oxidation state of manganese. $\left(Mn^{+4}, Mn^{+6}\right)$

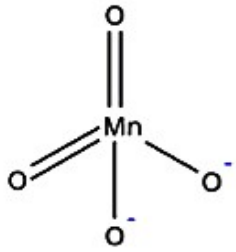
- **In reaction 3 & 4**, Cl_2 and O_3 used because they are good oxidising agent and oxidise Mn^{+6} to Mn^{+7}.
- So best method is electrolytic process to prepare $KMnO_4$.

  Properties: The above green solution is quite stable in alkali, but in pure water and in presence of acids, depositing MnO_2 and giving a purple solution of permanganate.

  $3K_2MnO_4 + 2H_2O \rightarrow \underset{\text{purple}}{2KMnO_4} + \underset{\text{dark brown}}{MnO_2\downarrow} + 4KOH$

Structure:





1.	Geometry	Tetrahedral	Tetrahedral
2.	Charge transfer	LMCT	d-d transition
3.	Bonding	$d\pi-p\pi$	$d\pi-p\pi$
4.	Colour	purple / pink	green
5.	Magnetic property	diamagnetic Paramagnetic (at high temperature) (at room temperature)	paramagnetic (at Room Temperature & high T)

- $KMnO_4$ act as oxidising agent in all medium

For above $E_1^\circ > E_2^\circ > E_3^\circ$

$$KMnO_4 \begin{cases} \xrightarrow{H^+} Mn^{2+} & E_1^\circ \\ \xrightarrow{H_2O/OH^-} MnO_2 & E_2^\circ \\ \xrightarrow{OH^-} MnO_4^{2-} & E_3^\circ \end{cases}$$

Example:

Oxidation in acidic medium in following reaction

In acidification of $KMnO_4$ only dilute H_2SO_4 is used. It oxidises as follows:

$$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$$

$$HCl \longrightarrow Cl_2$$

$$KMnO_4 \underset{(dil.\ H_2SO_4)}{\overset{(Acidified)}{+}} \left[\begin{array}{l} HBr \longrightarrow Br \\ HI \longrightarrow I_2 \\ HNO_2 \longrightarrow HNO_3 \\ H_3AsO_3 \longrightarrow H_3AsO_4 \\ H_2C_2O_4 \longrightarrow CO_2 \\ SO_2 \longrightarrow H_2SO_4 \\ H_2S \longrightarrow S \\ I_2 \longrightarrow HIO_3 \\ Fe^{2+} \longrightarrow Fe^{3+} \\ Sn^{2+} \longrightarrow Sn^{4+} \end{array}\right] + Mn^{+2}$$

- 1% alkaline $KMnO_4$ is called Baeyer's regent which is used to identify unsaturation of hydrocarbon.
- In $H_2C_2O_4$ vs $KMnO_4$ titration, $KMnO_4$ act as self indicator and Mn^{2+} is a self catalyst.
- $KMnO_4/H^+ + I^- \rightarrow Mn^{2+} + I_2$

  $KMnO_4 + I^- \rightarrow IO_3^- + MnO_2 + OH^-$

  $KMnO_4/H^+ + S_2O_3^{2-} \rightarrow S + SO_4^{2-} + Mn^{2+}$

  $KMnO_4 + S_2O_3^{2-} \rightarrow SO_4^{2-} + MnO_2 + OH^-$

Uses of $KMnO_4$:

Besides its use in analytical chemistry, potassium permanganate is used as a favourite oxidant in preparative organic chemistry. Its uses for the bleaching of wool, cotton, silk and other textile fibres and for the decolourisation of oils are also dependent on its strong oxidising power.

In laboratory conversion of Mn^{+2} to MnO_4^- is done by :

(i) PbO_2 (ii) $Pb_3O_4 + HNO_3$ (iii) $Pb_2O_3 + HNO_3$

(iv) $NaBiO_3/H^+$ (v) $(NH_4)_2S_2O_8/H^+$ (vi) KIO_4/H^+

Heating effect :

$$2KMnO_4 \xrightarrow[200°C]{\Delta} \underset{\text{green}}{K_2MnO_4} + \underset{\text{Black}}{MnO_2} + O_2$$

$$2K_2MnO_4 \xrightarrow[\text{hot}]{\text{at red}} K_2MnO_3 + O_2$$

Q10 **The yellow colour of chromates changes to orange on acidification due to the formation of-**

(A) Cr^{3+} **(B)** Cr_2O_3

(C) $Cr_2O_7^{2-}$ **(D)** CrO_4^-

Sol. C

$$\underset{\text{yellow}}{CrO_4^-} \rightleftharpoons \underset{\text{orange}}{Cr_2O_7^{-2}} \quad (H^+)$$

Q11 **Manganese show oxidation state from + 2 to + 7. The most oxidizing state known in aqueous solution is**

(A) + 7 **(B) + 4** **(C) + 3** **(D) + 2**

Sol. A

The most oxidizing state in aq. solution of Mn is +7 ($KMnO_4$)

Q12 **When conc. H_2SO_4 is added to $KMnO_4$, explosion take place and the product formed is-**

(A) Mn_2O_7 **(B)** MnO_2

(C) $MnSO_4$ **(D)** Mn_2O_3

Sol. A

When conc. H_2SO_4 is added to $KMnO_4$ explasion take block and Mn_2O_7 is formed.

Q13 **When $(NH_4)_2Cr_2O_7$ is subjected to heat, compound formed are-**

(A) NH_3, Cr_2O_3 **(B)** H_2O, $(NH_4)_2CrO_4$

(C) N_2, Cr_2O_3 **(D) All of these**

Sol. C

$$(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} NH_3 + Cr_2O_3 + H_2O$$

Different Compounds of D-block

1. **ZnO (Zinc oxide)**

It is also known as philosopher's wool due to its woolly flock type appearance.

Preparation:

By decomposition reaction of following substances

(i) $2Zn + O_2 \longrightarrow 2ZnO$

(ii) $ZnCO_3 \longrightarrow ZnO + CO_2$

(iii) $2Zn(NO_3)_2 \longrightarrow 2ZnO + 4NO_2 + O_2$

(iv) $Zn(OH)_2 \longrightarrow ZnO + H_2O$

Purest ZnO: To produced purest ZnO. Zinc sulphate treated with sodium carbonate, basic zinc carbonate precipitated and by heating only pure form of ZnO will produce along with gaseous by product.

Reaction

$$4ZnSO_4 + 4Na_2CO_3 + 3H_2O \longrightarrow \underset{\text{white basic zinc carbonate}}{ZnCO_3 \cdot 3Zn(OH)_2} \downarrow + 4Na_2SO_4 + 3CO_2$$

$$\xrightarrow{\text{heating}} 4ZnO + \underset{\text{pure}}{3H_2O} + CO_2$$

Chemical properties:

(i) $\underset{\text{white}}{ZnO(\text{cold})} \longrightarrow \underset{\text{yellow}}{Zn(\text{hot})}$

On heating, colour changes due to metal excess defects because some of oxide ions remove in form of

oxygen gas and leaves some electrons so that by movement of electrons in crystal colour changes.

(ii) It is insoluble in water

(iii) It sublime at 400°C.

(iv) It is amphoteric oxide.

$$ZnO + 2HCl \longrightarrow ZnCl_2 + H_2O$$

$$ZnO + H_2SO_4 \longrightarrow ZnSO_4 + H_2O$$

$$ZnO + 2NaOH \longrightarrow Na_2ZnO_2 + H_2O$$

Note:

All oxide and hydroxides are insoluble except NH_4^+, IA, Ba,Sr,Ca.

(v) We can obtain Zn by reduction with Carbon and H_2 gas.

$ZnO \longrightarrow Zn$ by H_2 and C

$ZnO + H_2 \longrightarrow Zn + H_2O$

$ZnO + C \longrightarrow Zn + CO$

(vi) If forms Rinmann's green with $Co(NO_3)_2$

$$2Co(NO_3)_2 \longrightarrow 2CoO + 4NO_2 + O_2$$

$$CoO + ZnO \longrightarrow \underset{\text{Rinmann's green}}{CoZnO_2} \text{ or } CoOZnO$$

Uses:

(i) As white pigment. It is superior than white lead because it does not turn into black

(ii) Rinmann's green is used as green pigment.

(iii) It is used as zinc ointment in medicine.

2. Zinc chloride, $ZnCl_2$

Preparation:

(i) Acid base reaction takes place in following

$$ZnO + 2HCl \longrightarrow ZnCl_2 + H_2O$$

$$ZnCO_3 + 2HCl \longrightarrow ZnCl_2 + H_2O + CO_2$$

$$Zn(OH)_2 + 2HCl \longrightarrow ZnCl_2 + 2H_2O$$

(ii) Anhydrous $ZnCl_2$ cannot be made by heating $ZnCl_2 \cdot 2H_2O$ because due to high polarising power of $ZnCl_2.2H_2O$.

$$ZnCl_2 \cdot 2H_2O \longrightarrow Zn(OH)Cl + HCl + H_2O$$

$$Zn(OH)Cl \longrightarrow ZnO + HCl$$

To get anhydrous $ZnCl_2$;

$$Zn + Cl_2 \longrightarrow ZnCl_2$$

$$Zn + 2HCl(dry) \longrightarrow ZnCl_2 + H_2$$

or $Zn + HgCl_2 \longrightarrow ZnCl_2 + Hg$

Properties

(a) It is deliquescent white solid (when anhydrous)

(b) $ZnCl_2 + H_2S \longrightarrow ZnS \downarrow$

$ZnCl_2 + NaOH \longrightarrow Zn(OH)_2 / Na_2[Zn(OH)_4]$

$ZnCl_2 + NH_4OH \longrightarrow Zn(OH)_2 / [Zn(NH_3)_4]^{2+}$

Uses:

(i) Used for impregnating timber to prevent destruction by insects.

(ii) As dehydrating agent when anhydrous,

(iii) $ZnO \cdot ZnCl_2$ used in dental filling.

3. Zinc sulphate, $ZnSO_4$:

Preparation:

$Zn + dil.H_2SO_4 \rightarrow ZnSO_4 + H_2$

$ZnO + dil.H_2SO_4 \rightarrow ZnSO_4 + H_2O$

$ZnCO_3 + dil.H_2SO_4 \rightarrow ZnSO_4 + H_2O + CO_2$

$ZnS + O_2 \rightarrow ZnO + SO_2$

$ZnS + 4O_3 \rightarrow ZnSO_4 + 4O_2$

$ZnSO_4 \cdot 7H_2O \rightarrow ZnSO_4 \cdot 6H_2O \rightarrow ZnSO_4 \cdot H_2O \rightarrow ZnSO_4 \rightarrow O_2 + SO_2 + ZnO$

Uses:

(i) In eye lotion.

(ii) Lithophone making ($ZnS + BaSO_4$) as white pigment.

Hydrated sulphate is of two types, one is $MSO_4 \cdot 5H_2O$

(Example :, $CuSO_4 \cdot 5H_2O$) and other is $MSO_4 \cdot 7H_2O$ Example :, $FeSO_4 \cdot 7H_2O$

4. Ferrous sulphate, $FeSO_4 \cdot 7H_2O$

Preparation:

(i) Scrap $Fe + H_2SO_4 \rightarrow FeSO_4 + H_2$

(ii) From kipp's waste $\overset{(dil.)}{}$

$FeS + H_2SO_4\,(dil.) \rightarrow FeSO_4 + H_2S$

(iii) $FeS_2 + 2H_2O + O_2 \rightarrow FeSO_4 + H_2SO_4$

Properties

(i) It undergoes aerial oxidation forming basic ferric sulphate

$$4FeSO_4 + H_2O + O_2 \rightarrow 4Fe(OH)SO_4$$

(ii) $FeSO_4 \cdot 7H_2O \rightarrow Fe_2O_3 + SO_2 + SO_3$

(iii) Aq. Solution is acidic due to hydrolysis

$$FeSO_4 + 2H_2O \rightarrow \underset{\text{Weak base}}{Fe(OH)_2} + H_2SO_4$$

(iv) It is a reducing agent.

(a) $Fe^{2+} + MnO_4^- + H^+ \rightarrow Fe^{3+} + Mn^{2+} + H_2O$

(b) $Fe^{2+} + Cr_2O_7^{2-} + H^+ \rightarrow Fe^{3+} + G^{3+} + H_2O$

(c) $Au^{3+} + Fe^{2+} \rightarrow Au + Fe^{3+}$

(d) $Fe^{2+} + HgCl_2 \rightarrow \underset{\text{White ppt.}}{Hg_2Cl_2} \downarrow + Fe^{3+}$

(v) It forms double salt. Example $(NH_4)_2SO_4 \cdot FeSO_4 \cdot 6H_2O$

5. Iron (II) oxide, FeO (black)

Preparation

$$FeC_2O_4 \rightarrow FeO + CO + CO_2$$

Properties

It is stable at high temperature and on cooling slowly disproportionate into Fe_3O_4 and iron

$$4FeO \rightarrow Fe_3O_4 + Fe$$

6. Iron(II) chloride, $FeCl_2$

Preparation:

$$Fe + 2HCl \rightarrow FeCl_2 + H_2$$

or

$$2FeCl_3 + H_2 \rightarrow 2FeCl_2 + 2HCl$$

Properties:

(i) It is deliquescent in air like $FeCl_3$

(ii) It is soluble in water, alcohol and ether also because it is sufficiently covalent in nature.

(iii) It volatilises at about 1000°C and vapour density indicates the presence of Fe_2Cl_4. Above 1300°C, density becomes normal.

(iv) It oxidises on heating in air.

$$12FeCl_2 + 3O_2 \rightarrow 2Fe_2O_3 + 8FeCl_3$$

(v) H_2 evolves on heating in stream

$$3FeCl_2 + 4H_2O \rightarrow Fe_3O_4 + 6HCl + H_2$$

(vi) It can exist in different hydrated form.

$FeCl_2 \cdot 2H_2O \rightarrow$ Colourless

$FeCl_2 \cdot 4H_2O \rightarrow$ pale green

$FeCl_2 \cdot 6H_2O \rightarrow$ green

7. Anhydrous $FeCl_3$

Preparation:

Anhydrous Ferric chloride is prepared by heating metallic iron in a stream of dry chlorine gas.

(i) $FeCl_3$ solid is almost black. IT sublines at about 300°C, giving a dimeric gas.

(ii) $FeCl_3$ dissolves in both ether and water, giving solvated monomeric species

(iii) Iron (III) chloride is usually obtained as yellow-brown lumps of the hydrate $FeCl_3 \cdot 6H_2O$

(iv) This is very soluble in water and is used both as an oxidizing agent, and as mordant in dyeing.

(v) $FeCl_3$ is also used in the manufacture of CCl_4.

Q14 **In the reaction $4Fe + 3O_2 \rightarrow 4Fe^{3+} + 6O^{2-}$; which of the following statement is incorrect -**

(A) It is a redox reaction **(B) Metallic iron is reduced to Fe^{3+}**

(C) Fe^{3+} is an oxidizing agent **(D) Metallic iron is a reducing agent**

Sol. B

Iron is oxidized

Q15 **Anhydrous ferric chloride is prepared by -**

(A) Dissolving ferric hydroxide in dilute HCl.

(B) Dissolving ferric hydroxide in conc. HCl.

(C) By passing dry Cl_2 gas over heated scrap iron

(D) By passing dry HCl gas over heated scrap iron

Sol. C

$2Fe + 3Cl_2 \rightarrow 2FeCl_3$

Q16 **Mohr's salt is a -**

(A) Normal salt **(B) Acid salt**

(C) Basic salt **(D) Double salt**

Sol. D

$(NH_4)_2 SO_4 . Fe SO_4 . 6H_2O$

8. CuO, (Copper oxide)

Preparation

(i) $CuCO_3 \cdot Cu(OH)_2 \rightarrow 2CuO + H_2O + CO_2$ (Commercial process)
Malachite Green
(native Cu-carbonate)

(ii) $2Cu + O_2 \rightarrow 2CuO$ and $Cu_2O + O_2 \rightarrow 2CuO$

(iii) $Cu(OH)_2 \rightarrow CuO + H_2O$

(iv) $2Cu(NO_3)_2 \rightarrow 2CuO + 4NO_2 + O_2$

Properties:

(i) CuO is insoluble in water

(ii) Readily dissolves in dil. Acids

$CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$

$CuO + HCl \rightarrow CuCl_2 + H_2O$

$CuO + HNO_3 \rightarrow Cu(NO_3)_2 + H_2O$

(iii) It decomposes when, heated above 1100°C

$4CuO \rightarrow 2Cu_2O + O_2$

(iv) CuO is reduced to Cu by H_2 or C under hot condition

$CuO + C \rightarrow Cu + CO$

$CuO + H_2 \rightarrow Cu + H_2O$

9. $CuCl_2$, (Cupric chloride):

Preparation:

$$CuO + 2HCl(conc.) \rightarrow CuCl_2 + H_2O$$

$$Cu(OH)_2 \cdot CuCO_3 + 4HCl \rightarrow 2CuCl_2 + 3H_2O + CO_2$$

Properties

(i) It is crystallised as $CuCl_2 \cdot 2H_2O$ of Emerald green colour.

(ii) dil. Solution in water is blue in colour due to formation of $\left[Cu(H_2O)_4\right]^{2+}$ complex.

(iii) Conc. HCl or KCl added to dil. Solution of $CuCl_2$ the colour changes into yellow, owing to the formation of $\left[CuCl_4\right]^{2-}$

(iv) The conc. Aq. Solution is green in colour having the two complex ions in equilibrium

$$2\left[Cu(H_2O)_4\right]Cl_2 \rightarrow \left[Cu(H_2O)_4\right]^{2+} + \left[CuCl_4\right]^{2-} + 4H_2O$$

(v) $CuCl_2 \rightarrow CuCl$ by no. of reagents

(a) $CuCl_2 + Cu$ – turnings $2CuCl$

(b) $2CuCl_2 + H_2SO_3 + H_2O \rightarrow 2CuCl + 2HCl + 2H_2SO_4$

(c) $2CuCl_2 + Zn/HCl \rightarrow 2CuCl + ZnCl_2$

(d) $CuCl_2 + SnCl_2 \rightarrow CuCl + SnCl_4$

10. $CuSO_4$,(Copper sulphate):

Preparation:

$$CuO + H_2SO_4(dil.) \rightarrow CuSO_4 + H_2O$$

$$Cu(OH)_2 + H_2SO_4(dil.) \rightarrow CuSO_4 + 2H_2O$$

$$Cu(OH)_2 \cdot CuCO_3 + H_2SO_4(dil.) \rightarrow CuSO_4 + 3H_2O + CO_2$$

$\underset{(Scrap)}{Cu} + H_2SO_4 + O_2 \rightarrow CuSO_4 + H_2O$ [commercial scale]

$Cu + dil.H_2SO_4 \rightarrow$ no reaction {Cu is a below H in electrochemical series}

Q17 **On heating $Cu(NO_3)_2$ strongly, the material finally obtained is-**

(A) Cu **(B) Cu_2O** **(C) $Cu(NO_2)_2$** **(D) $Cu(NO_3)_2$**

Sol. B

$$2Cu(NO_3)_2 \xrightarrow{\Delta} 2Cu_2O + 4NO_2 + O_2$$

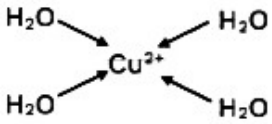
Q18 **In solid copper sulphate, copper is coordinated to -**

(A) Five water molecules **(B) four water molecules**

(C) One sulphate ion **(D) One water molecules**

Sol. B

In solid copper sulphate copper is coordinated to four water molecules



Compounds of silver

There are various compounds of silver which are made by treating silver with various compounds.

Preparation of various compounds :

Ag

- aq. regia → Not dissolved AgCl ↓
- Dry or Moist air → No effect
- H_2S → Black $\left[4Ag + 2H_2S + O_2 \rightarrow 2Ag_2S + 2H_2O\right]$
- H_2SO_4 → $2Ag + 2H_2SO_4 \rightarrow Ag_2SO_4 + SO_2 + H_2O$
- HNO_3 hot conc. → $Ag + 2HNO_3 \rightarrow AgNO_3 + NO_2 + H_2O$
- HCl → Not effected dil. HCl

$$E^\circ_{Ag/Ag^+} < E^\circ_{H_2/H^+}$$

Hence in presence of O_2

1. **Silver nitrate, $AgNO_3$**
   - It is called as lunar caustic because in contact with skin it produces burning like that of caustic soda with the formation of finely divided silver (black colour).
   - It is water soluble in aqueous medium and colourless solution
   - It is white in solid state.
2. **Silver sulfite, $Ag_2S_2O_3$**
   - It is whit solid precipitate.
   - It is soluble in excess $Na_2S_2O_3$ and form a complex and give black ppt of Ag_2S.

$$AgNO_3 + Na_2S_2O_3(aq.) \rightarrow \underset{\text{White}}{Ag_2S_2O_3} \downarrow$$

$$Ag_2S_2O_3 + Na_2S_2O_3 \rightarrow \left[Ag(S_2O_3)_2\right]^{3-}$$

$$Ag_2S_2O_3 \rightarrow \underset{\text{Black}}{Ag_2S} \downarrow + SO_4^{2-}$$

3. **Silver(I) oxide, Ag_2O**
   - It is black brown colour compound.

$$Ag_2CO_3(aq.) \xrightarrow{\Delta} \underset{\text{(Black-brown)}}{Ag_2O} \downarrow + CO_2 \uparrow$$

$$Ag_2O \xrightarrow{300^\circ C} Ag + O_2$$

   - Ag_2O is of basic nature.
   - Ag_2O acts oxidising agent.
   - Ag_2O is soluble in NH_3

Q19 **Zinc Oxide is-**

(A) A basic oxide **(B) An amphoteric oxide**

(C) An acidic **(D) A neutral**

Sol. B

It reacts with acid & base

Q20 **The compound which give off Oxygen on moderate heating is-**

(A) Cupric oxide **(B) Mercuric oxide**

(C) Zinc Oxide **(D) Aluminium oxide**

Sol. B

$2HgO \rightarrow 2Hg + O_2$

Q21 **Red precipitate is obtained when silver nitrate is added to -**

(A) K_2CrO_4 **(B) KI** **(C) KBr** **(D) $Na_2S_2O_3$**

Sol. A

$2AgNO_3 + K_2CrO_4 \rightarrow Ag_2CrO_4\downarrow + 2K^+ + 2NO_3^-$

Brick red ppt

Q22 **Preparation of looking mirror involve the use of-**

(A) Red lead

(B) Ammonical silver nitrate

(C) Ammonical $AgNO_3$ + Red lead

(D) Ammonical $AgNO_3$ + Red lead + HCHO

Sol. D

For the preparation looking mirror involve the use of $AgNO_3$ + Red lead + HCHO

Q23 **Which of the following forms with an excess of CN^-, a complex of coordination number two ?**

(A) Cu^{2+} **(B) Ag^+** **(C) Ni^{2+}** **(D) Fe^{2+}**

Sol. B

$[Ag(CN)_4]^-$ is a coordination complex of coordination no.2

Q24 **$AgNO_3$(s) produce black stain on the skin because of the reaction –**

(A) $2AgNO_3 \rightarrow 2Ag + 2NO_2 + O_2$ **(B) $2AgNO_3 \rightarrow Ag_2O + NO_2 + \frac{1}{2}O_2$**

(C) Both of these **(D) None of these**

Sol. A

$AgNO_3$(S) produce black stain on spin because of the reaction

$AgNO_3 \rightarrow 2Ag + 2NO_2 + O_2$

Black

F-Block

F-block have two series 4f and 5f named as Lanthanides and Actinides.

1. 4f-Series:

Lanthanoids (58-71) – Lanthanum(La) is also considered for properties in this series.

General Electronic configuration:

According to (n + l) rule ; configuration **[Xe] $4f^{1-14}$ $6S^2$**

Exception La , Ce & Gd (they have one-one e^- difference)

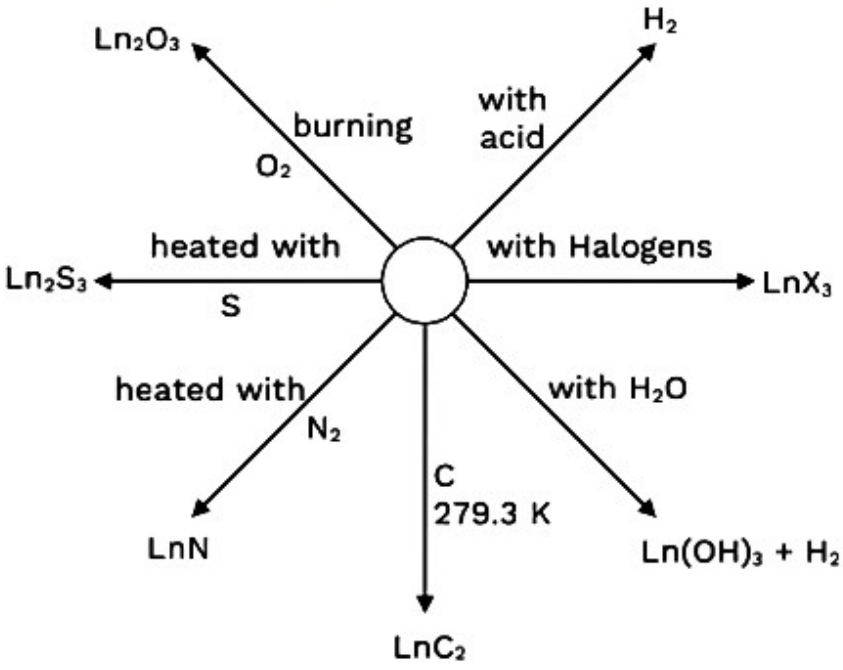
At. No.	Name	Symbol	E.C.
57.	Lanthanum	La	$5d^1 6s^2$
58.	Cerium	Ce	$4f^1 5d^1 6s^2$
59.	Praseodymium	Pr	$4f^3 6s^2$
60.	Neodymium	Nd	$4f^4 6s^2$
61.	Promethium	Pm	$5f^5 6s^2$
62.	Samarium	Sm	$4f^6 6s^2$
63.	Europium	Eu	$4f^7 6s^2$
64.	Gadolinium	Gd	$4f^7 5d^1 6s^2$
65.	Terbium	Tb	$4f^9 6s^2$
66.	Dysprosium	Dy	$4f^{10} 6s^2$
67.	Holmium	Ho	$4f^{11} 6s^2$
68.	Erbium	Er	$4f^{12} 6s^2$
69.	Thulium	Tm	$4f^{13} 6s^2$
70.	Ytterium	Yb	$4f^{14} 6s^2$
71.	Lutetium	Lu	$4f^{14} 5d^1 6s^2$

Occurrence - Lanthanoids are silvery soft metals found in earth's crust are called Rare earth element.

Properties:

1. **Reactivity** - Reactivity of Lanthanoids greater then hydrogen (Reactivity > H_2).
2. **Aqueous Solution** - f-block elements are generally coloured due to f-f transition except having configuration f^0, f^{14}.
3. **Oxidation state** - The common or stable oxidation state is +3 and also show +2 and +4 oxidation states
   +4 (O.N.) → act as oxidising agent
   +2 (O.N.)→ act as reducing agent

1. **Chemical reaction** -



Note:

Hydroxides of lanthanoides are basic in nature.

Uses:

(a) Lanthanoids are used to make alloy (95% Ln + % Fe = Mischmetal). Mischmetal is used to make bullets.

(b) Lanthanoids have property of phosphorescence.

(c) Lanthanoids are used to make TV screen.

(d) Lanthanoid oxides are used as catalyst in petroleum cracking.

2. 5f Series:

Actinoids (90-103) - All element of 5f series are radioactive.
Post uranium (92) elements starting from 93 are manmade elements.
General electronic configuration :

According to (n+l) rule- configuration $[Rn]5f^{14}6d^{0-12}7s^2$

Exception : following elements has exaptation of (n + l) rule (Thorium due to 2 - 2 electron difference).

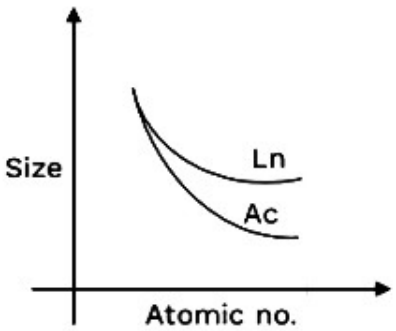
$2e^-$ difference

Ac (Th) Pa U Np Cm

Atomic no.	Name	Symbol	E.C.
89.	Actinium	Ac	$6d^17s^2$
90.	Thorium	Th	$6d^27s^2$
91.	Protactinium	Pa	$5f^26d^17s^2$
92.	Uranium	U	$5f^36d^17s^2$
93.	Neptunium	Np	$5f^46d^17s^2$
94.	Plutonium	Pu	$5f^67s^2$
95.	Americium	Am	$5f^77s^2$
96.	Curium	Cm	$5f^76d^17s^2$
97.	Berkelium	Bk	$5f^97s^2$
98.	Californium	Cf	$5f^{10}7s^2$
99.	Einsteinium	Es	$5f^{11}7s^2$
100.	Fermium	Fm	$5f^{12}7s^2$
101.	Mendelevium	Md	$5f^{13}7s^2$
102.	Nobelium	No	$5f^{14}7s^2$
103.	Lawrencium	Lr	$5f^{14}6d^17s^2$

Properties:

1. **Actinoid contraction :** Decrement in size of 5f series is due to poor shielding of 5f electrons is called Actinoid contraction. Actinoid contraction is more effective than lanthanoid contraction (4f).



2. **Oxidation states -** Actinoids shown more variety of oxidation state in comparison of lanthanoids this is due to energy gap in orbitals.

3. Energy gap 5f 6d 7s < 4f 5d 6s
Variation in oxidation state-

Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	No	Lr
+3		+3	+3	+3	+3	+3	+3	+3	+3	+3	+3	+3	+3
	+4	+4	+4	+4	+4	+4	+4	+4					
		+5	+5	+5	+5	+5							
			+6	+6	+6	+6							
				+7	+7								

- There is a greater range of oxidation states, which is in part attributed to the fact that the 5f, 6d and 7s levels of comparable energies.

4. **Reactivity** : reactivity of actinoids is greater than hydrogen.

Note:

Neodymium having high magnetic moment in 4f series which is explained by (spin only + orbital angular)

Q25 **Silver chloride is treated with aqueous sodium cyanide & the product thus formed is allowed to react with zinc in alkaline medium.**

Sol. AgCl dissolves in NaCN forming a complex sodium argento cyanide. The addition of Zn precipitates silver.

$AgCl + 2\ NaCN \rightarrow Na\ Ag\ (CN)_2 + NaCl$

$2Na\ Ag(CN)_2 + Zn \rightarrow K_2\ Zn\ (CN)_4 + 2Ag$

Q26 **Hot & concentrated nitric acid react with Copper.**

Sol. $Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$

Q27 **Cupric chloride is heated with concentrated HCl and Copper turnings.**

Sol. Cupric chloride is heated with conc. HCl and copper turnings –

$CuCl_2 + Cu \xrightarrow{HCl} Cu_2Cl_2$

Q28 **Potassium Iodide is added first and then Sodium thiosulphate to a solution of Copper Sulphate.**

Sol. $2CuSO_4 + 4KI \rightarrow Cu_2I_2 + 2K_2SO_4 + I_2$

$I_2 + 2Na_2S_2O_3 \rightarrow 2NaI + Na_2S_4O_6$

$2CuSO_4 + 4KI + 2Na_2S_2O_3 \rightarrow Cu_2I_2 + 2K_2SO_4 \rightarrow 2NaI + Na_2S_4O_6$

Q29 **Sulphur dioxide is circulated through a solution of copper sulphate containing sodium chloride.**

Sol. $2CuSO_4 + 2NaCl + SO_2 + 2H_2O \rightarrow Cu_2Cl_2 + Na_2\ SO_4 + 2H_2SO_4$

Q30 **Sodium thiosulphate solution is added to silver bromide.**

Sol. $AgBr + 2Na_2S_2O_3 \rightarrow Na_3[Ag(S_2O_3)_2] + NaBr$

sodium argentothio sulphate

Q31 **Excess of Caustic soda solution is added slowly to zinc chloride solution.**

Sol. $ZnCl_2 + 2NaOH \rightarrow Zn(OH)_2 + 2NaCl$

$Zn(OH)_2 + 2\,NaOH \rightarrow Na_2ZnO_2 + 2H_2O$

$ZnCl_2 + 4NaOH \rightarrow Na_2\,Zn\,O_2 + 2NaCl + 2H_2O$

Sodium zincate

Q32 **Excess of NH_4OH is added to $ZnCl_2$ solution.**

Sol. $ZnCl_2 + 2NH_4OH \rightarrow Zn(OH)_2 + 2NH_4Cl$

$Zn(OH)_2 + 2NH_4Cl + 2NH_4OH \rightarrow [Zn(NH_3)_4]\,Cl_2 + 4H_2O$

$ZnCl_2 + 4NH_4OH \rightarrow [Zn(NH_3)_4]\,Cl_2 + 4H_2O$

Tetrammine

zinc chloride

Q33 **Mercuric chloride is treated with NH_4OH.**

Sol. $HgCl_2 + 2NH_4OH \rightarrow Hg(NH_2)\,Cl + NH_4Cl + 2H_2O$

Q34 **KI is added to the excess solution of ferric sulphate.**

Sol. $Fe_2(SO_4)_3 + 2KI \rightarrow 2FeSO_4 + K_2SO_4 + I_2$

Q35 **Potassium permanganate solution is added to Ferrous Sulphate solution in presence of sulphuric acid.**

Sol. $2KMnO_4 + 3H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5\,[O]$

$2FeSO_4 + H_2SO_4 + [O] \rightarrow Fe_2(SO_4)_3 + H_2O \quad \times 5$

$2KMnO_4 + 8H_2SO_4 + 10FeSO_4 \rightarrow 5Fe_2(SO_4)_3 + K_2SO_4 + 2MnSO_4 + 8H_2O$

Q36 **H_2S is circulated through acidic solution of ferric Chloride.**

Sol. $2FeCl_3 + H_2S \rightarrow 2FeCl_2 + 2HCl + S$

Q37 **The first ionization potential of Copper is less than Zn while reverse is true for second ionization potential.**

Sol. $Cu \rightarrow 4s^1 \quad 3d^{10}$

$Zn \quad 4s^2 \quad 3d^{10}$

$Cu^+ \equiv 4s^2 \;\; 3d^{10} \;\; \& \quad Zn^+ = 4s^1 \;\; 3d^{10}$

Because Cu^+ has fully filled orbital which is stable configuration so the first I.E. of Cu in less. For removing one electron from Cu^+ fully filled orbital more energy has to be supplied that why second I.E. is high.

Q38 **Zinc readily liberates H_2 from cold dilute H_2SO_4 but not from cold conc. H_2SO_4.**

Sol. H_2SO_4 is a covalent compound conc. H_2SO_4 does not contain $H_3O^{\oplus}$ ions while dilute H_2SO_4 consists ions which react with zinc to liberate hydrogen

$H_2SO_4 + 2H_2O \rightleftharpoons 2\,H_3O^{\oplus} + SO_4^{-2}$

$Zn + 2\,H_3O^{\oplus} \rightarrow Zn^{2+} + H_2 + 2H_2O$

Q39 **Blue colour of $CuSO_4$ solution is discharged slowly when an iron rod is dipped into it.**

Sol. Cu^{2+} is reduced to Cu in presence of iron rod

$CuSO_4 + Fe \rightarrow Cu + FeSO_4$

Q40 **A small amount of sulphuric acid is added while preparing a standing solution of ammonium sulphate.**

Sol. $(NH_4)_2 SO_4 + 2H_2O \rightarrow 2\ NH_4OH + H_2SO_4$

ammonium sulphate is a salt of weak base and a strong acid. Its, hydrolysis occurs when it is dissolved in water and solution becomes turbid due to the formation of NH_4OH. Addition of small amount of acid shifts the equilibrium towards left and thus prevent hydrolysis.

Q41 **A solution of ferric chloride acidified with HCl is unaffected when hydrogen is bubbled through it, but get reduced when zinc is added to same acidified solution.**

Sol. Because of the formation of nascent hydrogen on addition of zinc. The solution of $FeCl_3$ is reduced.

Q42 **An aqueous solution containing one Mole of HgI_2 is orange in colour on addition of excess of NaI the solution become colourless. The orange colour reappear on subsequent addition of NaOCl. Explain with equation.**

Sol. $HgI_2 + 2NaI \rightarrow Na_2\ [Hg\ I_4]$

(Orange) coloured due to the presence of residued HgI_2

$HgI_2 + 2NaI$ (excess) $\rightarrow Na_2[HgI_4]$

(Orange) colourless

$3Na_2[HgI_4] + 2NaOCl + 2H_2O \rightarrow 2HgI_2 + 2NaCl + 4NaOH + 2NaI_3$

(orange)

Q43 **Which one of Fe^{2+} and Fe^{3+} ions is more paramagnetic and why?**

Sol. Fe^{3+} is more paramagnetic than Fe^{2+} as Fe^{3+} consists five unpaired electrons while Fe^{2+} possesses four unpaired electrons.

Q44 **Acidified $K_2Cr_2O_7$ solution turns green when sodium sulphite is added to it.**

Sol. Na_2SO_3 is a reducing agent. It reduces acidified $K_2Cr_2O_7$ to chromic sulphate which is green in colour.

$$K_2Cr_2O_7 + 4H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + 3[O]$$

$$\{Na_2SO_3 + [O] \rightarrow Na_2SO_4\} \times 3$$

$$K_2Cr_2O_7 + 2Na_2SO_3 + 4H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 3Na_2SO_4 + 4H_2O$$

green

Q45 **Zinc becomes dull in moist air.**

Sol. When zinc is exposed to moist air, the surface is affected with the formation of a film of basic zinc carbonate on it. Due to this zinc becomes dull.

$$[2Zn + 2H_2O + O_2 \rightarrow 2Zn(OH)_2] \times 2$$

$$Zn(OH)_2 + CO_2 \rightarrow ZnCO_3 + H_2O$$

$$ZnCO_3 + 3Zn(OH)_2 \rightarrow ZnCO_3 \cdot 3Zn(OH)_2$$

$$4Zn + 3H_2O + CO_2 + 2O_2 \rightarrow ZnCO_3 \cdot 3Zn(OH)_2$$

Q46 **A ferrous salt decolourises acidified $KMnO_4$ solution.**

Sol. Ferrous salt acts as a reducing agent it reduces acidified $KMnO_4$ in K_2SO_4 and $MnSO_4$ which form colourless solution, i.e. decoulorisation of $KMnO_4$ takes place.

$$2KMnO_4 + 10FeSO_4 + 8H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 5Fe_2(SO_4)_3 + 8H_2O$$

Q47 **Cadmium is used to extinguish nuclear fire.**

Sol. Cadmium absorbs neutrons, stops nuclear fission reaction and extinguishes nuclear fire

Q48 **Magnesium oxide is used for lining of steel making furnace.**

Sol. MgO is basic in nature. It removes acidic impurities present in cast iron used for making steels.

$MgO + SiO_2 \rightarrow MgSiO_3$ (slag)

$3MgO + P_2O_5 \rightarrow Mg_3(PO_4)_2$

MgO is also refractory meterial as it can tolerate very high temperature of the furnace

Q49 **Anhydrous $FeCl_3$ cannot be obtained by heating hydrated ferric chloride.**

Sol. Oh heating hydrated ferric chloride, anhydrous ferric chloride is not formed as water of cyrstallisation reacts to form Fe_2O_3 and HCl.

Q50 **Mercuric chloride is called corrosive sublimate.**

Sol. $HgCl_2$ is poisonous (corrosive) in nature. It undergoes sublimation due to these properties it is called corrosive sublimate.

Q51 **CuS is not precipitated by passing H_2S through copper sulphate solution containing KCN.**

Sol. $CuSO_4$ forms a complex with KCN

$CuSO_4 + 2KCN \rightarrow Cu(CN)_2 + K_2SO_4$

$2Cu(CN)_2 \rightarrow Cu_2(CN)_2 + (CN)_2$

$Cu_2(CN)_2 + 6KCN \rightarrow 2K_3[Cu(CN)_4]$

$\downarrow$

$6K^+ + 2(Cu(CN)_4)^{3-}$

$K_3[Cu(CN)_4]$ complex doe not furnish Cu^{2+} ions. Hence no ppt of CuS is formed when H_2S is passed through solution.

Complete & balance the following equations

Q52 $\mathbf{Fe_2O_3 . H_2O + NaOCl \rightarrow (A) + (B) + Cl_2}$

Sol. $Fe_2O_3.H_2O + NaOCl \rightarrow \underset{(A)}{2Na_2FeO_4} + \underset{(B)}{2HCl} + Cl_2$

Q53 $\mathbf{Fe_2O_3 + LiOH \xrightarrow{\Delta} C + H_2O}$

Sol. $Fe_2O_3 + 2LiOH \xrightarrow{\Delta} \underset{(C)}{2LiFeO_2} + H_2O$

Q54 $\mathbf{CuS_2O_3 + Na_2S_2O_3 \rightarrow D + E}$

Sol. $CuS_2O_3 + 2Na_2S_2O_3 \rightarrow 2NaCuS_2O_3 + Na_2S_4O_6$

Q55 $\mathbf{CuSO_4 + NH_4OH \rightarrow F + G}$

Sol. $CuSO_4 + 4\ NH_4OH \rightarrow Cu(NH_3)_4\ SO_4 + 4H_2O$

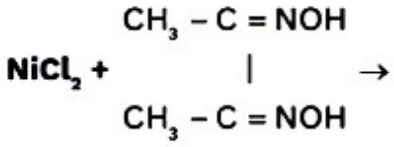
Q56 $\mathbf{MnO_4^- + Fe^{2+} \rightarrow Mn^{2+} + Fe^{3+} + H_2O}$

Sol. $MnO_4^- + 5Fe^{2+} + 8H^+ \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$

Q57 $\mathbf{K_2Cr_2O_7 + H_2SO_4 + SO_2 \rightarrow H + I + H_2O}$

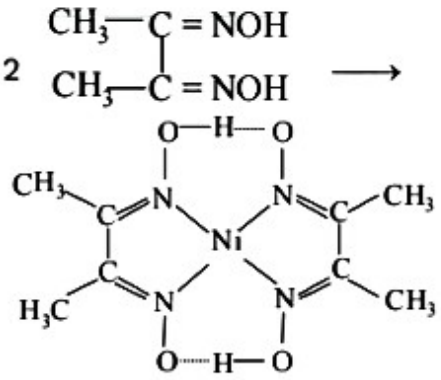
Sol. $K_2Cr_2O_7 + H_2SO_4 + 3SO_2 \rightarrow \underset{(H)}{K_2SO_4} + \underset{(I)}{Cr_2(SO_4)_3} + H_2O$

Q58



J + NH_4Cl + H_2O

Sol.



+ $2NH_4Cl + 2H_2O$

Q59 **$Cr_2O_7^{2-} + 2H^+ + H_2O_2 \rightarrow K + H_2O$**

Sol. $Cr_2O_7^{2-} + 2H^+ + 4H_2O_2 \rightarrow 2CrO(O_2)_2 + 5H_2O$

How will you obtain the following?

Q60 **Cuprous chloride from copper sulphate**

Sol. $2CuSO_4 + 2NaCl + SO_2 + 2H_2O \rightarrow \underset{\text{Cuprous Chloride}}{Cu_2Cl_2} + Na_2SO_4 + 2H_2SO_4$

Q61 **Silver from silver nitrate**

Sol. $2\,AgNO_3 + Zn \rightarrow 2\,Ag + Zn(NO_3)_2$

Q62 Cuprous oxide from copper sulphate

Sol. $CuSO_4 + 2NaOH \rightarrow \underset{\text{ppt.}}{Cu(OH)_2} + Na_2SO_4$

$$Cu(OH)_2 \xrightarrow[\text{heated}]{\text{strongly}} CuO + H_2O$$

$$CuO + Cu \text{ (powder)} \xrightarrow{\text{heat}} \underset{\text{Cuprous Oxide}}{Cu_2O}$$

Q63 Anhydrous $ZnCl_2$ from white vitriol

Sol. $ZnSO_4.7H_2O \xrightarrow[\text{heating}]{\text{strongly}} ZnSO_4 + 7H_2O$

$$ZnSO_4 \xrightarrow{\Delta} ZnO + SO_3$$

$$ZnO + 2HCl \rightarrow ZnCl_2 + H_2O$$

$$ZnCl_2 + NH_4Cl \rightarrow \underset{\text{Double Salt}}{ZnCl_2.2NH_4Cl}$$

$$ZnCl_2 . 2NH_4Cl \xrightarrow[HCl\ gas]{\text{Heat}} ZnCl_2 + 2NH_4Cl$$

Q64 Ferrous sulphate from ferrous ammonium sulphate.

Sol. $FeSO_4.(NH_4)_2SO_4 . 6H_2O + 4NaOH \rightarrow \underset{\text{ppt.}}{Fe(OH)_2} + 2NH_3 + 2Na_2SO_4 + 8H_2O$

$$Fe(OH)_2 + H_2SO_4 \rightarrow \underset{\text{Ferrous sulphate}}{FeSO_4} + 2H_2O$$

Q65 Gas (A) and Gas (B) both turns $K_2Cr_2O_7$ / H^+ green. Gas(A) also turns lead acetate paper black. When gas (A) is passed into gas (B) in aqueous solution, yellowish white turbidity appears. Identify gas(A) and (B) and explain reactions.

Sol. (A) : H_2S, oxidised to S by $Cr_2O_7^{2-}/ H^+$

(B) : SO_2 : oxidised to SO_4^{2-} by $Cr_2O_7^{2-} /H^+$

$$2H_2S + SO_2 \rightarrow 3S + 2H_2O$$

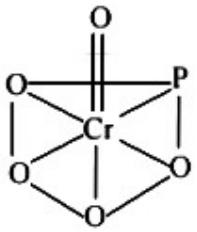
(yellowish white)

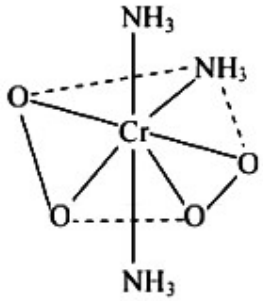
Q66 **(a) CrO_5 forms addition compound with pyridine in ethereal layer. What is structure of the addition product?**

(b) In ammoniacal solution, dichromate in presence of H_2O_2 forms $(NH_3)_3\ CrO_4$ in which Cr is + IV , the structure of compound is-

(c) CrO_3 is used as cleansing agent for glassware. How is it prepared?

(d) Give structure of $Cr_2O_7^{2-}$ ion.

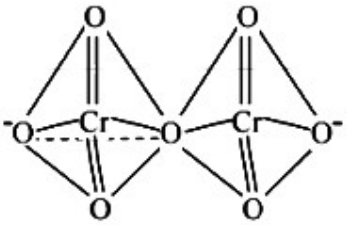
Sol. (a)  (Pentagonal pyramid)

(b)  Pentagonal bipyramid)

(c) $Na_2Cr_2O_7 + H_2SO_4 \rightarrow Na_2SO_4 + 2\ CrO_3 + H_2O$

Saturated Conc.

Aq.sol.

(d) 

Q67 **A certain inorganic compound (A) on heating loses its water of crystallization. On further heating a blackish brown powder (B) and two oxides of sulphur, (C) and (D) are obtained. The powder (B) on boiling with hydrochloric acid gives a yellow solution (E). When H_2S is passed in (E) a white turbidity (F) and a apple green colour (G) are obtained. The solution (E) on treatment with thiocyanate ions gives a blood red coloured compound (H). Identify from (A) to (H).**

Sol. (A) $FeSO_4 . 7H_2O$ (B) Fe_2O_3
(C) SO_2 (D)SO_3
(E) 2 $FeCl_3$ (F) S
(G) $FeCl_2$ (H) $Fe(SCN)_3$

Q68 **(a) A qualitative analysis test for Mn^{2+} involves its oxidation to MnO_4^- in acidic solution by sodium bismuthate, $NaBiO_3$. The bismuthate ion is reduced to Bi^{3+}. Write a balanced equation for this test.**
(b) What happens if PbO_2 is used instead of $NaBiO_3$?

Sol. (a) $5BiO_3^- + 2Mn^{2+} + 14H^+ \rightarrow 2MnO_4^- + 5Bi^{3+} + 7H_2O$
(b) $5PbO_2 + 2Mn^{2+} + 4H^+ \rightarrow 2MnO_4^- + 5Pb^{2+} + 2H_2O$

Q69 **(a) Colourless salt (A) gives white ppt. (B) with NaOH; ppt. (B) dissolves in excess of NaOH to form (C). On passing H_2S gas into (C) white ppt. (D) appears. (A) also gives white ppt. (E) with $AgNO_3$ solution. Identify (A) to (E).**
(b) What is the advantage of using ZnO in paints instead of Pb^{2+} salts.`
(c) Compare thermal stability of ZnO, CdO and HgO.

Sol. (a) (A) $ZnCl_2$ (B) $Zn(OH)_2$ (C) Na_2ZnO_2
(D) ZnS (E) AgCl
(b) ZnO in white paint by atmospheric H_2S gives Zn**S** (white) but Pb^{2+} salts in white paint turns black by atmospheric H_2S due to formation of PbS.
(c) ZnO > CdO > HgO

Q70 **A colourless salt (A) in dil. HCl gives black ppt. (B) on passing H_2S gas. ppt. (B) is insoluble in dil.HNO_3. (A) also gives white ppt. (C) on reaction with $AgNO_3$. Addition of KI in solution of (A) gives orange ppt. (D) which dissolves in excess of KI forming (E). Identify (A) to (D).**

Sol. (A) $HgCl_2$ (B) HgS (C) AgCl (D) HgI_2 (E) K_2HgI_4

Q71 **In acidic solution Silver (II) Oxide first dissolves to produce Ag^{2+}. This is followed by oxidation of H_2O to O_2 and the reduction of Ag^{2+} to Ag^+ . Write chemical equations for the dissolution and oxidation reduction reactions.**

Sol. Dissolution : $AgO + 2H^+ \rightarrow Ag^{2+} + H_2O$

Redox $4Ag^{2+} + 2H_2O \rightarrow 4\ Ag^+ + 4H^+ + O_2$

Q72 **Addition of F^- ions to a weakly acidic solution of $FeCl_3$ removes the colour, reduces the oxidizing power and interferes with the thiocyanate test for Fe^{3+} ion. Suggest an explanation.**

Sol. (A) : MnO_2

(B) K_2MnO_4

(C) $KMnO_4$

Q73 **Black coloured solid (A)** $\xrightarrow[\Delta]{KNO_3+KOH}$ **green coloured solution (B)**

$\downarrow CO_2$

(C) Pink + (A)

(C) is decolorized by Fe2+ Identify (A), (B) and (C) and explain reaction

Sol. F^- ion forms the complex ion, $[FeF_6]^{3-}$ This ion is colourless because this a weak-field complex (high spin). The d^5 configuration of Fe^{3+} has a spin forbidden transition and this ion also suffers form laporte and purity- forbidden transition. Thus, the ion is colourless, the ion cannot form the thiocyanato complex to give a red colour.

Q74 **The reduction potential for the Fe^{3+} /Fe^{2+} couple is + 0.76V in aqueous solution, it is reduced to – 0.12 V on the addition of excess of EDTA. Why?**

Sol. A reduction in electrode potential means a decrease in equilibrium constant because n FE = RTln K. In the absence of EDTA, log K = nFE/ 2.303 RT = nE/0.059 at 298 K . ($Fe^{3+} + e^- \rightarrow Fe^{2+}$). log K = 0.76/ 0.059, K = 7.6×10^{12}.

In absence of EDTA, log K = – 0.12/ 0.059 = – 2.033, K = 9.3×10^{-3}

This decrease can be explained on the basis of increase in entropy and chelate effects.

$Fe(H_2O)_6^{2+} + EDTA^{4-} \rightarrow \underset{\text{Stable}}{Fe(EDTA)^{2-}} + 6H_2O$

Fe^{3+} is rendered more stable by $edta^{4-}$ because of a spin crossover

Q75 **NO is a three - electron donor and CO is a two- electron donor. Using the E.A.N (18-electron) rule, find the values of x and y in the compounds $Cr(NO)_x$ and $Cr(CO)_y$.**

Sol. x = 4; y = 6

Q76 **Account for the subnormal magnetic moment (1.4 B.M.) per Cu atom of $[Cu(CH_3COO)_2]_2.2H_2O$.**

Sol. In the complex Cu^{2+} is a d^9 ion. It has one unpaired electron. According to spin -only formula, the complex must have a magnetic moment, $m_s = [1(1+2)]^{1/2} = 3^{1/2} = 1.732$ B.M. This is possible only if there is a weak interaction between the two Cu atoms. But if there is a firm Cu- Cu bond the complex would be diamagnetic.

Q77 **Dichromium acetate dihydrate, $Cr_2(OOCCH_3)_4 (OH_2)_2$, is diamagnetic though Cr^{2+} is a d^4 ion. Why ?**

Sol. In this complex, there is quadruple M-M bond between the two Cr atoms. The $d_{x^2-y^2}$ orbitals of the metal atoms are used for bonding to the ligands. The d_{z^2} atomic orbitals of the metal atom are used for bonding to the ligands. The d_{z^2} atomic orbital's form a σ – bond between the Cr atoms. d_{xz} and d_{yz} orbitals form two π- bonds. d_{xy} orbitals form a-(delta) bond. Because of this the complex is diamagnetic, even though free Cr^{2+} ion is a paramagnetic ion.